

Task-5: Insights & Summary

➤ **Summary:**

According to our predictive models, the most important properties that affect property valuation are spatial and structural assets, with total area, number of bathrooms and air conditioning being the top factors that most influence house prices. Both Linear Regression and Random Forest models successfully captured the key market trends and were able to predict the price of the properties with reliable precision; however the Random Forest model was much more powerful in predicting the price of more expensive properties during periods of non-linear pricing shifts. Interestingly, the traditional value-adding methods, such as central hot water, had only a slight positive effect on market value, and a completely unfurnished market had a much more negative impact on the final value of a property. The learning from these indicators and observations should shape real estates priorities towards structural footprint growth and modernization of the HVAC systems in places where more utility value can be gained for the least investment.

➤ **Which features influence house price the most?**

The mathematical correlation matrix and model weights corroborate that the most predictive leverage is given by Total Area (0.54) and Number of Bathrooms (0.52). While Air Conditioning (0.45) is a desirable extra, space and capacity are a property's basic, non-negotiable metrics, and are generally the first indicators used to determine its value

➤ **How accurate was my model?**

Both models are very accurate when used to estimate the normal market value of a firm in simple terms. The models' predictions of the market value of a house, which has traditional traits, are very close to the real value. The Random Forest model was found to be the best option for deployment because it was able to handle the multiple luxury features of the property (such as a multi-story layout with a modern AC), which increased the value of the property by an exponentially larger amount than simply by adding luxury features to the linear value.

➤ **What surprised me in the data?**

The most surprising conclusion was that Hot Water Heating (0.09) had virtually no independent relationship with price of a home. Traditional valuation approaches typically consider infrastructure improvements as premium value adders, the data however indicates that in the modern marketplace, infrastructure improvements are not considered as price drivers. Also, if the property does not have furniture listed, it will take a big hit with its Unfurnished designation (-0.28).

➤ **Recommendations:**

Real estate companies should look at acquiring features rather than renovate their structures, which can be costly. Data reveals that Air Conditioning and Parking have extremely high positive correlations with house prices (0.45 & 0.38 respectively). Agents should offer sellers to install modern and efficient split unit ACs and clearly define/pave parking spaces before listing. They're relatively inexpensive to implement, but have the greatest impact when it comes to the exact high-end luxury that buyers are willing to pay more for.